Date: EIOT Roll No.

ODD SEMESTER B.Tech (ECE, [redacted])

MID-SEMESTER EXAMINATION, SEPTEMBER 2025

Course Code- ECECC05/ EIECC05 Time: 1:30 Hrs

Course Title- Signals & Systems Max Marks: 25

Attempt all questions. Missing data/information (If any) may be suitably assumed & mentioned in the answer.

Q.No.		**Questions**	**Marks**	**CO**
Q 1	a	Given that x(t) and y(t) are periodic with time period equal to 'T'? Are the following signals periodic? If yes what is the time period. i. ax(t)+by(t), here a & b are constants. ii. x(t).y(2t) iii. x(mt)+y(nt), here m & n are integers.	3.5	CO1
	b	Compute the integral $\int_{-\infty}^{\infty}\{e^{-t}.\sum_{n=0}^{\infty}\delta(t-n)\}dt$	1.5	CO1
Q 2	a	Consider a discrete-time system whose input x[n] and output y[n] are related as $y[n]=\sum_{k=-\infty}^{\infty}2^{k-n}x[k+1]$. Is the system LTI? Is the system causal, stable?	3.5	CO1
	b	Determine whether x[n] = $\cos\left(\frac{\pi}{8}n^2\right)$ is an Energy or Power Signal.	1.5	CO1
Q 3	a	Compute and plot the convolution of x[n] and h[n]. Where, the input $x[n]=\left(\frac{1}{3}\right)^{-n}u[-n]$ and impulse response $h[n]=u[2-n]$. Compute the system output when the input is $x_2[n]=\left(\frac{1}{3}\right)^{1-n}u[1-n]$.	3.5	CO2
	b	y[n] = x[n+1] – 2x[n] + x[n-1]. Compute the impulse response h[n], and hence comment on the stability of the system.	1.5	CO1
Q 4	a	Let h(t) be the impluse response of a causal and stable continous time LTI system. Comment on the stability and causality of the cascade of two LTI sytems with impulse responses h(t) and h(-t).	3.5	CO2
	b	The unit step response of an LTI system is $s(t)=(1-e^{-2t})u(t)$. What is the response to an input $x(t)=4u(t+2)-u(t+1)$?	1.5	CO2
Q 5	a	Determine the fourier series coefficients of a periodic signal x(t) which has a time period 4. $x(t)=\begin{cases}-1, & -1\le t<0\\ 2, & 0\le t<2\\ -1, & 2\le t<3\end{cases}$	3.5	CO2
	b	Determine the DTFS coefficients of x(n)= 1+ cos(n π/12 +3 π/8)	1.5	CO2